

TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

Introduction

The Titanic Survival Prediction project aims to predict whether a passenger survived the Titanic disaster based on various features such as age, gender, passenger class, fare, and family details. This is a binary classification problem in machine learning.

Objective

To build a machine learning model that predicts passenger survival on the Titanic using historical passenger data.

Dataset

The dataset contains information about Titanic passengers, including:

- Passenger Class (Pclass)
- Gender (Sex)
- Age
- Fare
- Number of Siblings/Spouses (SibSp)
- Number of Parents/Children (Parch)
- Embarked Location
- Survival Status

Target Variable:

- Survived (0 = No, 1 = Yes)

Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Scikit-learn

Algorithm Used

Random Forest Classifier

Methodology

1. Load the Titanic dataset.
2. Handle missing values.
3. Remove unnecessary columns.
4. Convert categorical data into numerical form.
5. Split the dataset into training and testing sets.
6. Train the Random Forest model.
7. Predict survival outcomes.
8. Evaluate model performance using accuracy score.

Result

The Random Forest Classifier achieved an accuracy of 82.12% on the test dataset.

Accuracy Obtained:

82.12%

Conclusion

A machine learning model was successfully developed to predict Titanic passenger survival. The Random Forest algorithm produced an accuracy of 82.12%, demonstrating good predictive performance. The project highlights the importance of data preprocessing and feature selection in building effective classification models.